

ME 2261 - Embedded Systems Log Book

Memory Direct Addressing

```

1  .INCLUDE "M32DEF.INC" // Atmega32 microcontroller
2  .org $00
3
4  begin:LDI R25, 0x0B
5          STS 0x00A2, R25
6  // Program starts here
7
8  LDS R5, 0x00A2
9
10 END:NOP

```

Memory Indirect Addressing

<pre> 1 .INCLUDE "M32DEF.INC" // Atmega32 microcontroller 2 .org \$00 3 4 begin:LDI R26, 0x1C 5 LDI R27, 0x08 6 7 LDI R20, 0xFB 8 STS 0x081C, R20 9 // Program starts here 10 11 LD R11, X 12 13 END:NOP </pre>	<table border="0"> <tr><td>R04</td><td>0x00</td></tr> <tr><td>R05</td><td>0x00</td></tr> <tr><td>R06</td><td>0x00</td></tr> <tr><td>R07</td><td>0x00</td></tr> <tr><td>R08</td><td>0x00</td></tr> <tr><td>R09</td><td>0x00</td></tr> <tr><td>R10</td><td>0x00</td></tr> <tr><td>R11</td><td>0xFB</td></tr> <tr><td>R12</td><td>0x00</td></tr> <tr><td>R13</td><td>0x00</td></tr> <tr><td>R14</td><td>0x00</td></tr> </table>	R04	0x00	R05	0x00	R06	0x00	R07	0x00	R08	0x00	R09	0x00	R10	0x00	R11	0xFB	R12	0x00	R13	0x00	R14	0x00
R04	0x00																						
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R14	0x00																						

Memory Indirect Addressing with Displacement

<pre> 1 .INCLUDE "M32DEF.INC" // Atmega32 microcontroller 2 .org \$00 3 4 begin:LDI R28, 0x1C 5 LDI R29, 0x08 6 7 LDI R20, 0xFC 8 // Program starts here 9 10 STD Y+10, R20 11 12 END:NOP </pre>	Memory 1 Memory: prog FLASH <table border="1"> <tr><td>data 0x0826</td><td>fc 00 00 00 00</td></tr> <tr><td>data 0x083B</td><td>00 00 00 00 00</td></tr> <tr><td>data 0x0850</td><td>00 00 00 00 00</td></tr> <tr><td>data 0x0865</td><td>?? ?? ?? ?? ??</td></tr> <tr><td>data 0x087A</td><td>?? ?? ?? ?? ??</td></tr> <tr><td>data 0x088F</td><td>?? ?? ?? ?? ??</td></tr> <tr><td>data 0x08A4</td><td>?? ?? ?? ?? ??</td></tr> <tr><td>data 0x08B9</td><td>?? ?? ?? ?? ??</td></tr> </table>	data 0x0826	fc 00 00 00 00	data 0x083B	00 00 00 00 00	data 0x0850	00 00 00 00 00	data 0x0865	?? ?? ?? ?? ??	data 0x087A	?? ?? ?? ?? ??	data 0x088F	?? ?? ?? ?? ??	data 0x08A4	?? ?? ?? ?? ??	data 0x08B9	?? ?? ?? ?? ??
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data 0x08A4	?? ?? ?? ?? ??																
data 0x08B9	?? ?? ?? ?? ??																

Memory Indirect Addressing with Pre-decrement

```
1  .INCLUDE "M32DEF.INC" // Atmega32 microcontroller
2  .org $00
3
4  begin:LDI R26, 0x1C
5          LDI R27, 0x08
6
7          LDI R20, 0xAF
8  // Program starts here
9
10 ST -X, R20
11
12 END:NOP
```

Memory 1				
Memory:		prog FLASH		
data	0x081B	af	00	00 00
data	0x0830	00	00	00 00
data	0x0845	00	00	00 00
data	0x085A	00	00	00 00
data	0x086F	??	??	?? ??
data	0x0884	??	??	?? ??
data	0x0899	??	??	?? ??
data	0x08AE	??	??	?? ??

Memory Indirect Addressing with Post-increment

```
1  .INCLUDE "M32DEF.INC" // Atmega32 microcontroller
2  .org $00
3
4  begin:LDI R26, 0x1F
5          LDI R27, 0x07
6
7          LDI R20, 0xBA
8          MOV R15, R20
9  // Program starts here
10
11 ST X+, R15
12
13 END:NOP
```

Memory 1				
Memory:		prog FLASH		
data	0x071F	ba	00	00 00
data	0x0734	00	00	00 00
data	0x0749	00	00	00 00
data	0x075E	00	00	00 00
data	0x0773	00	00	00 00
data	0x0788	00	00	00 00
data	0x079D	00	00	00 00
data	0x07B2	00	00	00 00
data	0x07C7	00	00	00 00

Lab 05

AVR Assembler Programming (Addressing Modes)

Task 01

	R0	R1	R2	R3	R26	R27	X
clr r27	0	0	0	0	0	0	0
ldi r26, \$60	0	0	0	0	0x60	0	0x0060
ld r0, X+	0	0	0	0	0x61	0	0x0061
ld r1, X	0	0	0	0	0x61	0	0x0061
ldi r26, \$63	0	0	0	0	0x63	0	0x0063
ld r2, X	0	0	0	0	0x63	0	0x0063

```
1  .INCLUDE "M32DEF.INC" // Atmega32 microcontroller
2  .org $00
3
4  begin:CLR R27 // Clear R27
5      LDI R26, $60 // Load R26 with hexadecimal constant 60
6      LD R0, X+ // Load R0 with memory address X and increment X
7      LD R1, X // Load R1 with memory address X (X+1)
8      LDI R26, $63 // Load R26 with constant hexadecimal 63
9      LD R2, X // Load R2 with memory address X
10 // Program starts here
11
12 JMP 0x05
13
14 END:NOP
```

Task 02

	R0	R1	R2	R3	R4	R28	R29	Y
clr r29	0	0	0	0	0	0	0	0
ldi r28, \$60	0	0	0	0	0	0x60	0	0x0060
st Y+, r0	0	0	0	0	0	0x61	0	0x0061
ldi r28, \$63	0	0	0	0	0	0x63	0	0x0063
st Y, r2	0	0	0	0	0	0x63	0	0x0063
st -Y, r3	0	0	0	0	0	0x62	0	0x0062
std Y+2, r4	0	0	0	0	0	0x62	0	0x0062

```
1  .INCLUDE "M32DEF.INC" // Atmega32 microcontroller
2  .org $00
3
4  begin:CLR R29 // Clear R29
5  // Program starts here
6      LDI R28, $60 // Load R28 with hexadecimal constant 60
7      ST Y+, R0 // Store R0 in memory address Y and increment Y
8      ST Y, R1 // Store R1 in memory address Y (Y+1)
9      LDI R28, $63 // Load R28 with constant hexadecimal 63
10     ST Y, R2 // Store R2 in memory address Y
11     ST -Y, R3 // Decrement Y and Store R3 in memory address Y(Y-1)
12     STD Y+2, R4 // Store R4 in memory address Y+2
13
14 END:NOP
```

Task 03

Initial values in SRAM

Value	Address
\$34	\$0004
\$56	\$0003
\$44	\$0002
\$96	\$0001
\$00AA	\$07CD
\$00BB	\$07FF

	R1	R2	Applicable SRAM locations
lds r2, \$07FF	0x96	0xBB	\$0002, \$07FF
add r2, r1	0x96	0x51	\$0002, \$0001
sts \$07CD, r2	0x96	0x51	\$0002, \$07CD

```
1  .INCLUDE "M32DEF.INC" // Atmega32 microcontroller
2  .org $00
3
4  begin:
5      LDI R26, $5
6      LDI R16, $34
7      ST -X, R16
8      LDI R16, $56
9      ST -X, R16
10     LDI R16, $44
11     ST -X, R16
12     LDI R16, $96
13     ST -X, R16
14     LDI ZH, $07
15     LDI ZL, $CD
16     LDI R16, $AA
17     ST Z, R16
18     LDI R16, $BB
19     STD Z+50, R16
20     // Program starts here
21     LDS R2, $07FF // Load R2 with data of memory address $07FF
22     ADD R2, R1     // Add R2 and R1 and stored in R2
23     STS $07CD, R2 // Store R2 in memory address $07CD
24
25     END:NOP
```

Memory 1		
Memory:		prog FLA
data	0x07CD	51
data	0x07E2	00
data	0x07F7	00
data	0x080C	00
data	0x0821	00
data	0x0836	00
data	0x084B	00
data	0x0860	??
data	0x0875	??
data	0x088A	??
data	0x089F	??
data	0x08B4	??
data	0x08C9	??
data	0x08DE	??
data	0x08F3	??
data	0x0908	??
data	0x091D	??
data	0x0932	??
data	0x0947	??
data	0x095C	??
data	0x0971	??
data	0x0986	??
data	0x099B	??

Task 04

K =1; sum = 1;

While (k<=5)

{sum = sum*k;

k++; }

```
1  .INCLUDE "M32DEF.INC" // Atmega32 microcontroller
2  .org $00
3
4  begin:
5      LDI R16, 1      ; k=1
6      LDI R17, 1      ; sum=1
7      LDI R19, 5      ; constant 5 for comparison
8      // Program starts here
9  loop:
10     mov r18,r17 ; temp=sum
11     mul r18,r16 ; temp * k
12     mov r17,r0
13     clr r1
14
15     inc r16 ; k++
16
17     cpi r16, $6 ; compare k with 6
18     brge end_loop ; if k>= 6, exit loop
19     rjmp loop ; repeat loop
20 end_loop:
21
22  END:NOP
```

R16	0x06
R17	0x78
R18	0x18
R19	0x05